

? WinAnsiEncoding — Feature Overview

What this showcase demonstrates

TerraPDF ships with **extended WinAnsiEncoding width tables** covering the full byte range 32–255 — not just printable ASCII. Every accented character (Ä–ÿ), every Windows-1252 typographic special (€, …, ‘, ’, “, ”, —, —, ™) and every Latin-1 symbol is measured with its exact **Adobe AFM glyph width** so word-wrapping and justification stay pixel-perfect in any language.

Content-stream encoding: Non-ASCII characters are octal-escaped (\nnn) in PDF string literals, keeping the content stream pure ASCII while the reader resolves each byte via the font’s /WinAnsiEncoding vector. Metadata and bookmark titles use UTF-16BE hex strings (<FEFF…>) so viewer title bars and outline panels show the correct text too.

? Eighteen European Languages — Correct Rendering & Line-Wrapping

Language	Native Name	Sample Sentence (illustrates WinAnsi glyph coverage)
English	English	The quick brown fox jumps over the lazy dog.
French	Français	Le café au lait est servi avec des croissants frais.
German	Deutsch	In der Münchener Straße öffnete das Schloß früh.
Spanish	Español	El señor Martínez cruzó la montaña en cañón.
Portuguese	Português	O coração não mente — é a essência da nação.
Italian	Italiano	La città di Venezia è più bella al tramonto.
Dutch	Nederlands	Ögenschiijnlijk is het zoiets als déjà-vu.
Romanian	Romana	Soarele răsare în fiecare dimineată peste dealuri.
Hungarian	Magyar	Gyula és Ágnes sétáltak a városban, ahol a fák zöldek.
Norwegian	Norsk	Bjørn gikk på fjordstien i Ålesund om kvelden.
Swedish	Svenska	Björk och Åsa åkte till Göteborg på lördag.
Finnish	Suomi	Yöllä tähtitaivas heijastuu järven ylle.
Polish	Polski	Na starym wzgórzu stoi zamek, a niebo jest modre i jasne.
Czech	Cestina	Praha je zlaté srdce Evropy — slavná, stará a nádherná.
Turkish	Türkçe	Türk kültürü zengin ve güzel — müzik, dans ve sanat bir arada.
Catalan	Català	El Barça jugà a Girona i guanyà per sis a zero.

TerraPDF — Unicode & WinAnsiEncoding Showcase

European Language Coverage

Language	Native Name	Sample Sentence (illustrates WinAnsi glyph coverage)
Danish	Dansk	Søren Ørsted stod på broen og så på åen.
Welsh	Cymraeg	Mae'r iâ ar y mynydd yn toddi yn yr haf.

All 18 sentences rendered using Helvetica/Times via WinAnsiEncoding — glyph widths sourced from Adobe Core-14 AFM tables.

? Windows-1252 Specials (byte range 0x80–0x9F)

Unicode code points U+0080–U+009F are C1 controls and are never mapped to WinAnsi. Instead, bytes 0x80–0x9F encode 27 distinct characters — each with its own Unicode code point and AFM advance width.

Unicode	WinAnsi	Glyph	Name	Example in context
U+2018	0x91	‘	Left single quotation mark	‘Hello’, he said.
U+2019	0x92	’	Right single quotation mark	It’s a fine day.
U+201C	0x93	“	Left double quotation mark	“She smiled.”
U+201D	0x94	”	Right double quotation mark	Said “merci”.
U+2013	0x96	—	En dash	Pages 42–47.
U+2014	0x97	—	Em dash	Time—and tide—wait.
U+2026	0x85	...	Horizontal ellipsis	To be continued...
U+2022	0x95	•	Bullet	• First item
U+20AC	0x80	€	Euro sign	Price: € 1,299.00
U+2122	0x99	™	Trade mark sign	TerraPDF™
U+0152	0x8C	Œ	OE ligature (capital)	Œuvre complète
U+0153	0x9C	œ	OE ligature (small)	Manœuvre
U+2020	0x86	†	Dagger	See note † below.
U+2021	0x87	‡	Double dagger	Also ‡ ibid.
U+2030	0x89	‰	Per mille sign	Error rate: 1.2‰
U+0160	0x8A	Š	S with caron (capital)	Škoda Auto
U+0161	0x9A	š	S with caron (small)	Dušan

? Typographic Specials in Running Prose

Curly quotes: ‘Single’ and “double” quotation marks look far more professional than straight ASCII apostrophes.

Dashes & ellipsis: An en–dash separates page ranges (pp. 42–47). An em—dash sets off a clause—without spaces. An ellipsis... trails off elegantly.

Symbols: Copyright © 2025, registered trademark ®, trade mark ™. Price: € 1?299.00. Temperature: 23°C ± 0.5°. $3^2 + 4^2 = 5^2$ (Pythagorean triple). Rate: 2.4‰ per mille.

Ligatures & AE: The Œuvre of Æschylus — manœuvre, nœud, æon — all rendered from WinAnsiEncoding glyphs.

? Latin-1 Supplement Symbol Groups (U+00A0–U+00FF)

Group	Characters
Punctuation & Spacing	« » ¿ ¡ · ¶ §
Mathematical & Technical	± × ÷ ° μ ² ³ ¼ ½ ¾
Currency & Commerce	¢ £ ¤ ¥ ¦ © ® ¯
Accented Capitals (À–Ö)	À Á Â Ã Ä Å Æ Ç È É Ê Ë Ì Í Î Ï Ð Ñ Ò Ó Ô Õ Ö
Accented Capitals (Ø–þ)	Ø Ù Ú Û Ü Ý Þ ß
Accented Lowercase (à–ö)	à á â ã ä å æ ç è é ê ë ì í î ï ð ñ ò ó ô õ ö
Accented Lowercase (ø–ÿ)	ø ù ú û ü ý þ ÿ

TerraPDF — Unicode & WinAnsiEncoding Showcase

? WinAnsiEncoding — Complete Byte-to-Glyph Reference (0x20–0xFF)

Every cell shows the rendered glyph, its decimal byte value, and Unicode code point. Grey cells are undefined positions. Bytes 0x80–0x9F are Windows-1252 specials; 0xA0–0xFF are Latin-1 Supplement.

	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
0x20	• 32 U+0020	! 33 U+0021	" 34 U+0022	# 35 U+0023	\$ 36 U+0024	% 37 U+0025	& 38 U+0026	' 39 U+0027	(40 U+0028	) 41 U+0029	* 42 U+002A	+ 43 U+002B	, 44 U+002C	- 45 U+002D	. 46 U+002E	/ 47 U+002F
0x30	0 48 U+0030	1 49 U+0031	2 50 U+0032	3 51 U+0033	4 52 U+0034	5 53 U+0035	6 54 U+0036	7 55 U+0037	8 56 U+0038	9 57 U+0039	: 58 U+003A	; 59 U+003B	< 60 U+003C	= 61 U+003D	> 62 U+003E	? 63 U+003F
0x40	@ 64 U+0040	A 65 U+0041	B 66 U+0042	C 67 U+0043	D 68 U+0044	E 69 U+0045	F 70 U+0046	G 71 U+0047	H 72 U+0048	I 73 U+0049	J 74 U+004A	K 75 U+004B	L 76 U+004C	M 77 U+004D	N 78 U+004E	O 79 U+004F
0x50	P 80 U+0050	Q 81 U+0051	R 82 U+0052	S 83 U+0053	T 84 U+0054	U 85 U+0055	V 86 U+0056	W 87 U+0057	X 88 U+0058	Y 89 U+0059	Z 90 U+005A	[91 U+005B	\ 92 U+005C	] 93 U+005D	^ 94 U+005E	_ 95 U+005F
0x60	` 96 U+0060	a 97 U+0061	b 98 U+0062	c 99 U+0063	d 100 U+0064	e 101 U+0065	f 102 U+0066	g 103 U+0067	h 104 U+0068	i 105 U+0069	j 106 U+006A	k 107 U+006B	l 108 U+006C	m 109 U+006D	n 110 U+006E	o 111 U+006F
0x70	p 112 U+0070	q 113 U+0071	r 114 U+0072	s 115 U+0073	t 116 U+0074	u 117 U+0075	v 118 U+0076	w 119 U+0077	x 120 U+0078	y 121 U+0079	z 122 U+007A	{ 123 U+007B	 124 U+007C	}	~ 125 U+007D	— 126 U+007E
0x80	€ 128 U+20AC	—	, 130 U+201A	f 131 U+0192	” 132 U+201E	… 133 U+2026	† 134 U+2020	‡ 135 U+2021	^ 136 U+02C6	% 137 U+2030	Š 138 U+0160	‹ 139 U+2039	Œ 140 U+0152	—	Ž 142 U+017D	—
0x90	—	‘ 145 U+2018	’ 146 U+2019	“ 147 U+201C	” 148 U+201D	• 149 U+2022	— 150 U+2013	— 151 U+2014	~ 152 U+02DC	™ 153 U+2122	š 154 U+0161	› 155 U+203A	œ 156 U+0153	—	ž 158 U+017E	ÿ 159 U+0178
0xA0	• 160 U+00A0	ı 161 U+00A1	¢ 162 U+00A2	£ 163 U+00A3	¤ 164 U+00A4	¥ 165 U+00A5	¦ 166 U+00A6	§ 167 U+00A7	¨ 168 U+00A8	© 169 U+00A9	ª 170 U+00AA	« 171 U+00AB	¬ 172 U+00AC	­ 173 U+00AD	® 174 U+00AE	¯ 175 U+00AF
0xB0	° 176 U+00B0	± 177 U+00B1	² 178 U+00B2	³ 179 U+00B3	´ 180 U+00B4	µ 181 U+00B5	¶ 182 U+00B6	· 183 U+00B7	¸ 184 U+00B8	¹ 185 U+00B9	º 186 U+00BA	» 187 U+00BB	¼ 188 U+00BC	½ 189 U+00BD	¾ 190 U+00BE	¿ 191 U+00BF
0xC0	À 192 U+00C0	Á 193 U+00C1	Â 194 U+00C2	Ã 195 U+00C3	Ä 196 U+00C4	Å 197 U+00C5	Æ 198 U+00C6	Ç 199 U+00C7	È 200 U+00C8	É 201 U+00C9	Ê 202 U+00CA	Ë 203 U+00CB	Ì 204 U+00CC	Í 205 U+00CD	Î 206 U+00CE	Ï 207 U+00CF
0xD0	Ð 208 U+00D0	Ñ 209 U+00D1	Ò 210 U+00D2	Ó 211 U+00D3	Ô 212 U+00D4	Õ 213 U+00D5	Ö 214 U+00D6	× 215 U+00D7	Ø 216 U+00D8	Ù 217 U+00D9	Ú 218 U+00DA	Û 219 U+00DB	Ü 220 U+00DC	Ý 221 U+00DD	Þ 222 U+00DE	ß 223 U+00DF
0xE0	à 224 U+00E0	á 225 U+00E1	â 226 U+00E2	ã 227 U+00E3	ä 228 U+00E4	å 229 U+00E5	æ 230 U+00E6	ç 231 U+00E7	è 232 U+00E8	é 233 U+00E9	ê 234 U+00EA	ë 235 U+00EB	ì 236 U+00EC	í 237 U+00ED	î 238 U+00EE	ï 239 U+00EF
0xF0	ð 240 U+00F0	ñ 241 U+00F1	ò 242 U+00F2	ó 243 U+00F3	ô 244 U+00F4	õ 245 U+00F5	ö 246 U+00F6	÷ 247 U+00F7	ø 248 U+00F8	ù 249 U+00F9	ú 250 U+00FA	û 251 U+00FB	ü 252 U+00FC	ý 253 U+00FD	þ 254 U+00FE	ÿ 255 U+00FF

- Printable ASCII (0x20–0x7E)
- Win-1252 Specials (0x80–0x9F)
- Latin-1 Supplement (0xA0–0xFF)
- Undefined

All 224 defined WinAnsiEncoding positions (32–255) — each glyph measured with its Adobe AFM advance width.